

CS222: Computer Organization and Architecture

Mid Sem, Spring 2014

Attempt all questions

Time: 2 Hrs

Full marks: 30

1. Find the binary representation of -169.5 in IEEE 754 single precision format. (3)
2. Convert the C function below to ARM assembly language. Make sure that your assembly language code could be called from a standard C program.

```
unsigned int sum(unsigned int n) {  
    if (n == 0) return 0;  
    else return n + sum(n-1);  
}
```

[Register convention for ARM, r0-r3 - Argument registers (not preserved), r4-r11 - Variable registers (preserved), sp - Stack pointer (preserved), lr - Link register (preserved), r12 - Scratch register for inter-procedure call (not preserved)] (6)

3. Describe an algorithm for sequential version of multiplication to compute the product of two 32 bit unsigned numbers. Present a working example to demonstrate your algorithm on 4-bit numbers. (2.5+2.5) (6)
4. Describe a pipelined datapath needed to support only 'sw rd, offset(rs)' instruction. Your datapath should be designed with the assumption that only this type of instruction will be executed. (6)
5. Explain with example data hazard in Arithmetic instructions in the context of MIPS pipeline architecture. How such thing can be detected and avoided? (10)